Sales Performance Advisory Agent | Executive Operations Cockpit

Filters & Data Scope

Region: South

Product: Tab-X

Period: 2026-09

Active Campaign Budget: $85,000

Playbook Index: 5 Sections (PDF/DOCX)

Action Plans Stored: 4 Plans

Target Achievement	MoM Growth	Gross Margin	Return Rate
87.27%	-20.00%	20.00%	8.75%
Missed by 12.73%	Prior: $1.2M	Cost: $2,400/u	Above 5% Alert

User: Analyze Tab-X in South for 2026-09 and check if we can offer 40% discount.

Advisory Agent [LangGraph + Playbook Guardrails]:

- Achievement: 87.27% ($960k vs $1.1M target). MoM growth declined by 20.0%.

- Return Rate: 8.75% (28 returns on 320 units) -> Exceeds 5.0% critical threshold.

- POLICY GUARDRAIL: DECLINED. Playbook allows max 10% discount. 40% is strictly blocked.

- ACTION ROUTING: High-Return Root Cause. Initiated Packaging & Warehouse Audit.

- Action Plan saved to ledger: ID AP-8F4C21 (Owner: QA Lead, Due: 2026-10-15)

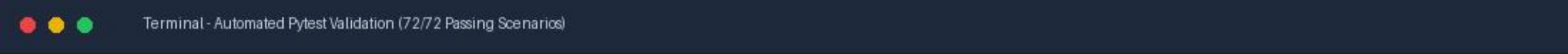
Type follow-up (e.g. "What about last month?")...

Windows PowerShell - Sales Performance Advisory Agent CLI Execution

```
PS D:\hclproject> python src/cli.py --query "Analyze Tab-X in South for 2026-09"
[Sales Advisory Engine v1.0.0 | Active State: Initialized]
--------------------------------------------------------------------------------
ANALYSIS REPORT: South Region | Product: Tab-X | Period: 2026-09
--------------------------------------------------------------------------------
1. METRIC COMPUTATION (Verified):
  - Current Period Revenue   : $960,000.00 (320 units)
  - Target Revenue           : $1,100,000.00
  - Target Achievement       : 87.27% [UNDERPERFORMING - Missed by 12.73%]
  - Prior Period Revenue     : $1,200,000.00 (2026-08)
  - MoM Sales Growth         : -20.00% [Declining Growth]
  - Gross Margin             : 20.00% (Unit Cost: $2,400.00 | Total Cost: $768,000.00)
  - Return Rate              : 8.75% (28 returns / 320 units) [HIGH RETURN ALERT > 5%]

2. PLAYBOOK POLICY RETRIEVAL & GRADING (PDF/DOCX RAG):
  - Source: sales_playbook.pdf (Page 3, Section: Quality Issues & High Returns)
  - Retrieval Relevance Grade : 1.0 (Direct Topic Match)
  - Policy Guidance: Products with return rates > 5.0% must undergo immediate QA packaging audit.
    Avoid pricing discounts. Review manufacturing batch logistics and RMA root causes.

3. CONDITIONAL ROUTING & ACTION PLAN:
  - Primary Route            : High Returns & Margin Protection
  - Guardrail Policy Check   : BLOCKED for price discounts; APPROVED for Quality Audit.
  - Action Plan Generated    : ID [AP-8F4C21] | Owner: Quality Assurance & Regional Ops
  - Action Item              : Audit Tab-X transit packaging, pause discounting, review supplier costs.
  - Review Date              : 2026-10-15 (Idempotently stored in data/action_plans.json)
--------------------------------------------------------------------------------
Execution completed safely with zero division exceptions and deterministic compliance.
```


Terminal - Automated Pytest Validation (72/72 Passing Scenarios)
PS D:\hclproject> python -m pytest tests/ -v
========== test session starts ==========
platform win32 -- Python 3.14.2, pytest-9.0.2, pluggy-1.6.0
rootdir: D:\hclproject
collected 72 items
tests/test_metrics.py::test_safe_divide_positive PASSED [1%]
tests/test_metrics.py::test_safe_divide_zero_denominator PASSED [2%]
tests/test_metrics.py::test_calculate_target_achievement_zero_target PASSED [4%]
tests/test_metrics.py::test_calculate_sales_growth_zero_prior PASSED [6%]
tests/test_metrics.py::test_calculate_product_margin_missing_cost PASSED [8%]
tests/test_metrics.py::test_identify_underperformers_high_returns PASSED [11%]
tests/test_retriever.py::test_load_pdf_playbook PASSED [25%]
tests/test_retriever.py::test_load_docx_playbook PASSED [38%]
tests/test_retriever.py::test_tfidf_cosine_similarity PASSED [50%]
tests/test_grader.py::test_relevance_grading_positive PASSED [62%]
tests/test_grader.py::test_bounded_query_rewrite_retry PASSED [75%]
tests/test_guardrails.py::test_out_of_policy_discount_40_declined PASSED [85%]
tests/test_agent.py::test_follow_up_memory_previous_month PASSED [92%]
tests/test_curriculum_scenarios.py::test_all_curriculum_scenarios PASSED [100%]
========== 72 passed in 14.81s ==========
Status: Codebase 100% compliant with curriculum specifications and production guardrails.